

Why Rotating Fluids Make Polygons

Paper III: Gravitational Theory

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Abstract

Building on the vortex–gravity isomorphism established in Paper II, we develop the gravitational theory of polygon stability in 2+1 and higher dimensions. The orbifold CFT at central charge $c = 12b(N) \approx N^2$ identifies the Havelock eigenvalue with the twisted-sector ground-state energy (the orbifold–Havelock correspondence). The polygon–BTZ phase transition has critical exponent $\nu = 1/2$, with entropy decomposing into discrete (wall-crossing, $\log 2$ per threshold) and continuous (one-loop) parts. The Wheeler–DeWitt equation on the palindromic minisuperspace connects the stability threshold to the Hamiltonian constraint. The main result: in 2+1D, the Havelock eigenvalue structure—applied via the equivalence principle (test polygons at every point of a spatial slice)—*derives* the vacuum Einstein equations $R_{\mu\nu} = \Lambda g_{\mu\nu}$: entropy maximisation of the frozen polygon modes on a general background forces constant curvature, through three hypotheses (H1 from the $\csc^2$ uniqueness theorem, H2 from entropy maximisation of the frozen polygon modes, H3 from the Hadamard universality of the Green’s function). In $d \geq 3$, the characterisation extends (the Havelock condition is equivalent to vacuum Einstein, providing a new criterion for Einstein manifolds). The Bernoulli number $B_2 = 1/6$ organises the spectral structure (Paper I): the stability polynomial $6B_2(x) = P_2(2x-1)$ places the threshold at the Gauss–Legendre quadrature nodes, and the $\csc^2$ kernel connects the vortex spectrum to integrable systems from random matrix theory to gauge theory.

1 Introduction and summary

Paper II establishes that N equal point masses on a spatial slice of 2+1D gravity interact via the same Green’s function as point vortices (Proposition 10.1), giving identical Havelock eigenvalues $\lambda_m = C_1(\xi) - m(N-m)/2$. The conformal invariance of the flat-plane backreaction (Proposition 10.2) ensures that $N_{\text{crit}} = 7$ is independent of the deficit angle—a remarkable simplification that makes the algebraic-number thresholds of Paper I into gravitational phase transitions.

This paper develops the consequences:

- §2–§3: the graviton at $N = 7$ and the spectral bound as a quantum gravitational bound.
- §4–§5: holographic interpretation and the conformal block identification.
- §6: the polygon–BTZ phase transition, entropy accounting, and Lorentzian continuation.

- §7–§8: the Wheeler–DeWitt equation and the three-layer decomposition.
- §9–§10: higher-dimensional extension and triangle universality.
- §11–§12: the gravity partition function and the derivation of the Einstein equations.

Logical architecture of the central charge. A single quantity $c = 12b(N)$ appears in four roles in this paper: as the WDW mass parameter (§7), the CS level $k = c/6$ (§2), the orbifold CFT central charge (§8), and the Todd-class correction in the equivariant index (§8, Proposition 8.2). These are *not four independent claims*; they form a single logical chain:

1. *Definition (classical).* The three-layer decomposition of the Havelock eigenvalue $\lambda_m = C_1 - f_m + \delta_m$ extracts a unique mode-independent offset $b(N)$ (Proposition 8.2). Define $c := 12b(N)$. This is a *classical* computation: $b(N)$ is the unique value making the Toeplitz determinants of the csc^2 kernel positive for all angular modes (a total-positivity condition on the circulant matrix, requiring no quantum input).
2. *The WDW kinetic coefficient.* The Feynman–Kac mapping of the Onsager partition function (§12, Proposition 12.2) produces a kinetic coefficient in the WDW equation. This coefficient is $c = 12b(N)$ — the same c defined in step 1 from the eigenvalue data. No gravitational input is required: c is determined by the vortex spectrum alone.
3. *Consistency check (Brown–Henneaux).* Defining $k = c/6$ and $G = 3/(2c)$, the Brown–Henneaux relation $c = 3\ell/(2G)$ is *automatically satisfied* (it reduces to $c = c$ after substitution). This confirms that the c extracted from the eigenvalues is the boundary Virasoro central charge, but this identification is a *consequence*, not an input.
4. *Orbifold matching.* At leading order $c \approx N^2$, the correspondence with the $\mathbb{Z}_N$ orbifold CFT (Proposition 8.1) follows from $c/(24N^2) \approx 1/2$. The $O(N)$ discrepancy between c and N^2 is the regularisation of the vortex self-energy.

Every subsequent use of c in this paper traces back to step 1 (extraction from the eigenvalue data). The gravitational interpretation (Brown–Henneaux) is a consistency check at step 3, not a logical prerequisite.

2 The graviton at $N = 7$ and the Chern–Simons lift

The stability boundary $N_{\text{crit}} = 7$ has a representation-theoretic explanation. In the CS formulation of 2+1D gravity (Witten 1988), the vortex polygon is a Wilson line whose eigenvalue is the $\text{SL}(2, \mathbb{R})$ Casimir $C_2(R) = j(j+1)$ of the representation R (Paper II). The Havelock Casimir $f(m, N) = m(N-m)/2$ IS this $\text{SL}(2, \mathbb{R})$ Casimir—not by coincidence, but because the vortex eigenvalue structure IS the CS Wilson line spectrum. The critical mode has $f(m^*, N) = j(j+1)$ with integer j only at specific N :

N	m^*	$f(m^*, N)$	j such that $j(j+1) = f$
4	2	2	$j = 1$ (vector boson)
7	3	6	$j = 2$ (graviton)

No other $N \leq 30$ gives an integer spin at the critical mode. The *graviton* (spin-2 field of general relativity) appears precisely at $N = 7$: the Havelock stability threshold is the onset of the gravitational degree of freedom. The total csc^2 sum $\lambda_0 = \sum_{p=1}^{N-1} \text{csc}^2(\pi p/N) = (N^2-1)/3$ itself has the Casimir form $\lambda_0 = (4/3)j(j+1)$ with $j = (N-1)/2$, where $4/3 = 1 + 2B_2$ (Paper I, §??). The same $4/3$ that governs the self-interaction of the csc^2 kernel also sets the normalisation of the total angular stiffness of the polygon.

Proposition 2.1 (Graviton identification). *$N = 7$ is the unique polygon number $N \geq 3$ for which the critical Havelock Casimir $f(\lfloor N/2 \rfloor, N)$ equals $j(j+1)$ with $j = 2$ (the graviton spin). Higher Pell solutions ($N = 41, j = 14$; $N = 239, j = 84$; ...) give $j > 2$ and lie outside the stability range $N \leq 7$.*

Proof. For even $N = 2k$: $f(k, 2k) = k^2/2$. Then $j(j+1) = k^2/2$ gives $j = (-1 + \sqrt{1 + 2k^2})/2$. This is an integer iff $1 + 2k^2$ is a perfect square, i.e. $n^2 - 2k^2 = 1$ (Pell equation). Solutions: $(n, k) = (1, 0), (3, 2), (17, 12), \dots$. No even N yields $j = 2$: at $k = 2$ ($N = 4$), $j = 1$; at $k = 12$ ($N = 24$), $j = 8$. Setting $j = 2$ requires $n = 5$, giving $25 - 2k^2 = 1$, i.e. $k^2 = 12$, which has no integer solution.

For odd $N = 2k+1$: $f(k, 2k+1) = k(k+1)/2$. Then $j(j+1) = k(k+1)/2$ gives $j = (-1 + \sqrt{1 + 2k(k+1)})/2$. At $k = 3$ ($N = 7$): $j(j+1) = 6$, $j = 2$. At $k = 4$ ($N = 9$): $j(j+1) = 10$, $j \approx 2.70$ (not integer). At $k = 5$ ($N = 11$): $j(j+1) = 15$, $j \approx 3.41$. For $4 \leq k \leq 19$: j is irrational (the next Pell solution is $k = 20$, $N = 41$, $j = 14$). Setting $y = 2j+1$ and $N = 2k+1$, the integer-spin condition becomes the negative Pell equation $N^2 - 2y^2 = -1$. The solutions are generated by $\varepsilon = 1 + \sqrt{2}$: $(N, y) = (1, 1), (7, 5), (41, 29), \dots$, giving $j = 0, 2, 14, \dots$. The first non-trivial solution is $N = 7$, $j = 2$. $\square$

The Chern–Simons lift. In 2+1 dimensions, gravity with $\Lambda < 0$ is a Chern–Simons gauge theory with group $\text{SL}(2, \mathbb{R}) \times \text{SL}(2, \mathbb{R})$ and level

$$k = \frac{\ell}{4G} = \frac{c}{6}, \quad (2.1)$$

where the Brown–Henneaux relation $c = 6k$ gives $k = 2b(N)$ for the exact central charge $c = 12b(N)$ (with $b(N) = N(N+1)/12 - \ln 2 + (\ln N)/(N-1)$, Paper I, verified to 10^{-16} for $N = 3, \dots, 24$). At leading order $c \approx N^2$, so $k \approx N^2/6$. The CS level $k = c/6$ is not integer for most N . Level quantisation in CS theory arises from large gauge transformations; for the non-compact $\text{SL}(2, \mathbb{R})$, the relevant topological constraint is weaker than for compact groups [?, §3]. Level quantisation requires integer k only for *compact* groups ($\pi_3(G) = \mathbb{Z}$); for $\text{SL}(2, \mathbb{R})$, $\pi_3 = 0$ and no topological obstruction exists [?, §3]. The compact gauge sectors ($\text{SU}(2)_1, \text{SU}(3)_1$) have integer levels; only the gravitational $\text{SL}(2, \mathbb{R})$ sector carries the irrational level $k = 2b(N)$, which is well-defined for the non-compact group. The vortex polygon is a *Wilson line* $W_m = \text{Tr}_{R_m} \mathcal{P} \exp \oint A$ in the representation R_m with Casimir $f(m, N)$. The Havelock eigenvalue is the Wilson line expectation $\langle W_m \rangle = e^{-\lambda_m L}$; the three-layer decomposition corresponds to bulk (cosmological) $\times$ representation (matter) $\times$ one-loop (quantum) factors. The CS partition function on Seifert manifolds is computed Stieltjes–Wigert polynomials [?], linking the Havelock eigenvalue sequence to the log-normal moment problem.

(The Havelock matrix is a circulant on $\mathbb{Z}/N\mathbb{Z}$ with the log-sine kernel; the Casimir $m(N-m)/2$ is the Fourier eigenvalue and δ_m the lattice correction.)

3 The spectral bound as a quantum gravitational bound

The spectral correction and the gravitational one-loop partition function are both moments of the heat kernel:

$$\log Z_{\text{grav}} = - \int_0^\infty [\text{Tr } e^{-t\Delta_S} - 1] \frac{dt}{t} \quad (\text{weight } 1/t), \quad (3.1)$$

$$\delta C_1(z_0) = \int_0^\infty [K(z_0, z_0, t) - K_{\mathbf{H}^2}(t)] dt \quad (\text{weight } 1). \quad (3.2)$$

The first is the functional determinant $\det \Delta_S$, the second determines polygon stability. Both involve the Selberg zeta function $Z_\Gamma(s)$ at the zero $s = 1$, but this requires care.

The zero at $s = 1$. The Selberg zeta $Z_\Gamma(s)$ has zeros at s such that $s(1-s)$ is an eigenvalue of Δ_S . The constant eigenfunction ($\lambda_0 = 0$) gives a zero at $s = 1$, so $Z_\Gamma(1) = 0$ —not a finite number. The Euler product $\prod_\gamma \prod_k (1 - e^{-(s+k)\ell_\gamma})$ converges for $\text{Re}(s) > 1$ but diverges at $s = 1$ (by the prime geodesic theorem: the partial product over geodesics with $\ell \leq L$ decays to 0 as $L \rightarrow \infty$).

The physically relevant quantities are the *residue* at this zero:

- $Z'_\Gamma(1)$: the *regularised* determinant $\det' \Delta_S = \prod_{n \geq 1} \lambda_n$ (omitting $\lambda_0 = 0$). This is the gravitational one-loop partition function [Giombi et al., 2008].
- The spectral sum $\sum_{n \geq 1} |\varphi_n(z_0)|^2 / \lambda_n$: the regularised Green's function at the diagonal, which determines $\delta C_1(z_0)$.

Both are finite and well-defined; both involve the analytic structure of Z_Γ at its zero, not the (divergent) Euler product.

The spectral computation of δC_1 . At the Aut-fixed centre z_0 of the Bolza surface, only trivial-representation eigenfunctions contribute: $\delta C_1(z_0) = \sum_{n: \text{triv}} |\varphi_n(z_0)|^2 / \lambda_n \approx 0.011$, dominated by the first trivial-rep eigenvalue $\lambda_{\text{triv}} \approx 15.05$ (the first eigenvalue in the trivial representation of $\text{Aut}(\text{Bolza})$; the full spectral gap is $\lambda_1 \approx 3.84$; Strohmaier and Uski 2013), with rapid convergence: $1/(4\pi \cdot 15.05) \approx 0.005$, $1/(4\pi \cdot 23.08) \approx 0.003$, $1/(4\pi \cdot 32.67) \approx 0.002$, The spectral bound is rigorous: the ratio of consecutive trivial-representation eigenvalues $\lambda_n / \lambda_{n+1} \leq 15.05 / 23.08 = 0.65$ (from the Strohmaier–Uski data; Weyl asymptotics guarantee $\lambda_n \sim cn$ for large n , so the ratio decreases) gives $\sum_{n \geq 1} 1/\lambda_n \leq (1/\lambda_1)/(1-0.65) = 2.86/\lambda_{\text{triv}}$, so $|\delta C_1| \leq 2.86/(4\pi \lambda_{\text{triv}}) \approx 0.015$.

The spectral gap λ_1 of the Laplacian controls both quantities. In the gravitational language:

- $C_1(\mathbf{H}^2, \xi)$: classical stability (flat-space AdS, no topology).
- δC_1 : one-loop quantum correction (from the topology of the spatial slice).
- The bound $|\delta C_1| \leq 2.86/(4\pi \lambda_{\text{triv}})$ at the Aut-fixed centre ($\lambda_{\text{triv}} \approx 15.05$ for the Bolza surface), or $|\delta C_1| \leq 2.86/(4\pi \lambda_1)$ at a generic point ($\lambda_1 \approx 3.84$): quantum gravitational corrections to polygon stability.

The Selberg dictionary. Four distinct physical quantities involve the Selberg zeta at the zero $s = 1$:

Z_Γ data at $s = 1$	Physical meaning
$Z'_\Gamma(1)$ (regularised determinant $\det' \Delta_S$)	Gravitational partition function $= \prod_{n \geq 1} \lambda_n$
$\sum_{n \geq 1} \varphi_n(z_0) ^2 / \lambda_n$ (spectral; convergent)	Vortex stability correction δC_1 Localised at z_0 ; Aut-suppressed
Palindromic threshold polynomials (from $C_1(\xi) = f_{\max}(N)$)	Stability transitions: values of ξ where N_{crit} jumps
Hecke eigenvalues of associated forms (a_p from modular forms)	Prime-by-prime visibility Which thresholds are visible on $\Gamma(p)$

The first two rows involve the same zero $s = 1$ of Z_Γ but differ in what they extract: the determinant (a *product* of eigenvalues) versus the resolvent trace (a *sum* of $1/\lambda_n$, localised at z_0). Neither can be computed from the Euler product (which diverges at $s = 1$); both require either the spectral data or the analytic continuation of Z_Γ .

The palindromic thresholds control *where* stability transitions occur (at which curvature ξ); the eigenvalue spectrum controls *how large* the correction δC_1 is. On highly symmetric surfaces (the modular surface, the Bolza surface), the systole and the palindromic geodesics coincide; on a generic arithmetic surface they are distinct.

4 Holographic interpretation

In $\text{AdS}_3/\text{CFT}_2$, a point mass m in the bulk creates a conical deficit that corresponds to the insertion of a heavy operator $\mathcal{O}$ of conformal dimension $h = (c/24)(1 - (1 - 8Gm)^2)$ in the boundary CFT [Carlip, 1998]. A regular N -gon of equal masses at radial coordinate r in the bulk corresponds to N equally-spaced heavy operator insertions on the boundary circle.

In this language, the Havelock decomposition takes the form:

Bulk (2+1D AdS gravity)	Boundary (1+1D CFT)
Perturbation mode m	$\mathbb{Z}_N$ conformal block channel
$f(m, N) = m(N-m)/2$	Casimir of the m -th channel
$C_1(\xi)$	Function of h/c and AdS radius
$\lambda_m = C_1 - f(m, N) > 0$	Positivity of the m -th block
$C_1 \geq f_{\max}(N)$	Bootstrap-type bound on h/c

The stability condition $C_1 \geq m(N-m)/2$ for all m becomes a *lower bound on the conformal dimension* of the boundary operators: below this bound, the bulk polygon is unstable and the corresponding $\mathbb{Z}_N$ conformal block changes sign. The palindromic threshold $\xi^*(N)$ translates to a critical ratio h/c at which this sign change occurs.

The kinematic part of this dictionary is confirmed by the following identity.

Proposition 4.1 (Havelock Casimir = one-chiral OPE Casimir). *Let $T_m = \sum_{p=1}^{N-1} \frac{1 - \cos(2\pi pm/N)}{2 \sin^2(\pi p/N)}$ be the $\mathbb{Z}_N$ Fourier eigenvalue of the OPE kernel $\csc^2(\pi p/N)$. Then*

$$T_m = m(N-m) = 2 f(m, N), \quad (4.1)$$

exactly twice the Havelock Casimir $f(m, N) = m(N-m)/2$.

Proof. Write $\omega = e^{2\pi i/N}$ and $S_m = \sum_{p=1}^{N-1} (1 - \omega^{pm})/|1 - \omega^p|^2$. Since $|1 - \omega^p|^2 = 2(1 - \cos(2\pi p/N)) = 4\sin^2(\pi p/N)$ and $1 - \omega^{pm} = 1 - \cos(2\pi pm/N) - i\sin(2\pi pm/N)$, the real part gives $T_m = \text{Re}(S_m) = \sum_{p=1}^{N-1} [1 - \cos(2\pi pm/N)]/(2\sin^2(\pi p/N))$. Now consider the polynomial identity on $\mathbb{Z}/N\mathbb{Z}$: the function $f_m(p) = [1 - \cos(2\pi pm/N)]/(2\sin^2(\pi p/N))$ extends to a trigonometric polynomial of degree $\leq N-1$. Its sum equals the coefficient of z^0 in the Laurent expansion of $\frac{1-z^m}{1-z} \cdot \frac{1-z^{-m}}{1-z^{-1}}$, evaluated at $z = \omega$ and summed over $p = 1, \dots, N-1$. This product is $|\sum_{k=0}^{m-1} z^k|^2$, whose z^0 coefficient (from the orthogonality $\sum_{p=0}^{N-1} \omega^{pm} = N\delta_{n,0}$) is $\min(m, N-m) \cdot \max(m, N-m) = m(N-m)$, since exactly m terms contribute in each sector. $\square$

The factor of 2 has physical content. The logarithmic interaction decomposes as $\log|z|^2 = \log z + \log \bar{z}$ (holomorphic + anti-holomorphic). In the boundary CFT, the OPE of two heavy operators has both chiral sectors, giving the *full* Casimir $m(N-m)$. The Havelock eigenvalue uses $f = m(N-m)/2$, the contribution of *one* chiral sector, because:

- The Kirchhoff equations $\dot{z}_k = -i\partial H/\partial \bar{z}_k$ are first-order: coordinates are conjugate to each other (no separate momenta). This halves the phase-space dimension, selecting one chiral sector of the $\log|z|^2 = \log z + \log \bar{z}$ interaction.
- The stability criterion $\lambda_m > 0$ is $C_1 > m(N-m)/2$: the confining potential must exceed the holomorphic OPE repulsion in the m -th $\mathbb{Z}_N$ channel.

In gravity (second-order dynamics), both sectors contribute to the equations of motion, but the stability criterion—positivity of the Hessian eigenvalue—still involves $m(N-m)/2$ per sector (Proposition 10.1).

The remaining step is to express C_1 in terms of boundary CFT data. On the flat plane, $C_1 = N-1$ (the Havelock self-energy); on $\mathbf{H}^2$ at curvature parameter ξ ,

$$C_1(\xi) = (N-1) \frac{1+\xi^2}{(1-\xi)^2}. \quad (4.2)$$

The curvature parameter ξ encodes the radial position of the bulk polygon; through the Brown–Henneaux relation $c = 3\ell/(2G)$ and the deficit angle–dimension map $h = (c/24)(1-(1-8Gm)^2)$, the stability condition becomes a bound on conformal dimensions parametrised by the bulk position. The palindromic thresholds $\xi^*(N)$ give algebraic-number bulk positions at which the stability bound is saturated:

N	$\xi^*(N)$	d_{crit}/ℓ	$\varepsilon = 1 - \xi^*$	h/c_{max}	Field
8	$8-3\sqrt{7}$	0.51	0.937	0.018	$\mathbb{Q}(\sqrt{7})$
10	$1/7$	0.80	0.857	0.015	$\mathbb{Q}$
12	$(18-5\sqrt{11})/7$	0.97	0.798	0.013	$\mathbb{Q}(\sqrt{11})$
23	φ^{-2}	1.44	φ^{-1}	0.007	$\mathbb{Q}(\sqrt{5})$

Here d_{crit}/ℓ is the geodesic distance from the AdS centre in units of the AdS radius, and $\varepsilon = 1 - \xi^*$ is the boundary proximity (the polygon must sit at $\varepsilon < \varepsilon^*(N)$, i.e., sufficiently close to the AdS boundary, to be stable). The $N = 23$ threshold gives $\varepsilon^* = \varphi^{-1} = (\sqrt{5}-1)/2$: the golden ratio appears as a critical bulk-position parameter in the holographic dictionary.

The two constraints on the polygon are independent:

- *Mass*: $h/c < h/c_{\max}(N)$ (total deficit $< 2\pi$; from the bulk geometry).
- *Position*: $\xi > \xi^*(N)$ (confining potential $>$ OPE repulsion; from the Havelock decomposition and the chiral identity (4.1)).

Together, they define the allowed region in the $(h/c, \xi)$ parameter space for a stable N -gon of heavy operators—a two-parameter family of bootstrap-type bounds, one for each N , all descending from the single identity $\lambda_m = C_1 - m(N-m)/2$.

The confining potential from the OPE kernel. The formula $C_1 = (N-1)(1+\xi^2)/(1-\xi)^2$ can be derived either from the Kimura Hamiltonian for point vortices on $\mathbf{H}^2$ (Paper I, §4) or from the second radial variation of the chordal OPE kernel $\sigma(z, w) = |z-w|/|1-\bar{z}w|$ at the N -gon. The two derivations are mathematically identical: both compute the Hessian of $-\sum_{j<k} \log \sigma(z_j, z_k)$ on the Poincaré disk. The factor $(1+\xi^2)/(1-\xi)^2$ is *not* the conformal factor $\Omega(\sqrt{\xi})^2 = 1/(1-\xi)^2$ alone; the additional $(1+\xi^2)$ comes from the angular structure of σ under radial perturbations.

Relation to the conformal bootstrap. The stability condition $\lambda_m > 0$ and the conformal bootstrap positivity conditions (crossing symmetry + unitarity, [Rattazzi et al. 2008](#)) are both positivity conditions on conformal block coefficients, but in *different channels*:

	Bootstrap	Vortex stability
What is constrained	CFT spectrum $\{\Delta_i\}$	Bulk polygon position ξ
Input	c, Δ_{gap}	$N, h/c$
Positivity condition	OPE $f_{ijk}^2 \geq 0$	$\lambda_m = C_1 - f(m, N) > 0$
Channel	generic 4-point OPE	$\mathbb{Z}_N$ -symmetric

The two are *complementary*: the bootstrap constrains which CFTs exist, while the vortex stability constrains which bulk configurations are stable in a given CFT. The identification $\lambda_m \leftrightarrow$ OPE coefficient would make the stability condition a bootstrap condition for $\mathbb{Z}_N$ -symmetric correlators—a direction requiring the $\mathbb{Z}_N$ conformal block computation sketched below.

5 The $\lambda_m \leftrightarrow$ OPE identification

The Havelock eigenvalue is entirely computable from boundary CFT data. The key is to decompose the semiclassical N -point function $\langle \mathcal{O}(z_1) \cdots \mathcal{O}(z_N) \rangle$ of N equally-spaced heavy operators into $\mathbb{Z}_N$ channels: the angular channels give $m(N-m)/2$ (the kinematic OPE Casimir, Proposition 4.1), and the radial confinement gives $C_1(\xi)$.

In the semiclassical (large- c) limit, the N -point function factorises as

$$\log \langle \mathcal{O}(z_1) \cdots \mathcal{O}(z_N) \rangle \approx -2h \sum_{j<k} \log \sigma(z_j, z_k) + O(1), \quad (5.1)$$

where $\sigma = |z_j - z_k|/|1 - \bar{z}_j z_k|$ is the chordal distance. The leading term is the “free OPE energy”; the $O(1)$ correction comes from the Virasoro block (graviton exchange in the bulk).

Under *angular* perturbation $z_k \rightarrow R e^{i(2\pi k/N + \delta\theta_k)}$ at the N -gon, the second variation of the leading term decomposes into $\mathbb{Z}_N$ Fourier modes $\delta\theta_k = \sum_m a_m \cos(2\pi mk/N)$:

$$\delta^2 \left[-2h \sum_{j < k} \log \sigma \right] = 2h \sum_{m=1}^{N-1} m(N-m) |a_m|^2. \quad (5.2)$$

Proposition 5.1 (Angular OPE stiffness). *The coefficient of $|a_m|^2$ in (5.2) is $2h \cdot m(N-m) = 2h \cdot T_m$, where $T_m = m(N-m)$ is the full OPE Casimir (Proposition 4.1). This follows analytically from Proposition 4.1 ($T_m = m(N-m)$, proved) and the constrained second variation; verified numerically for $N \leq 12$.*

The missing piece is the confining potential $C_1(\xi)$. We now show it too arises from the boundary.

Theorem 5.2 ($\mathbb{Z}_N$ conformal block = Havelock eigenvalue). *Let $\mathcal{O}_1, \dots, \mathcal{O}_N$ be N identical heavy operators of dimension h , placed at equally-spaced points $z_k = \sqrt{\xi} e^{2\pi i k/N}$ on the boundary of the Poincaré disk. In the semiclassical limit ($c \rightarrow \infty$, h/c fixed):*

- (a) Angular stiffness. *The m -th $\mathbb{Z}_N$ channel of the angular second variation of the N -point function gives $2h \cdot m(N-m)$ (Proposition 5.1).*
- (b) Radial confinement. *The log-radial ($u = \log \sqrt{\xi}$) second variation of $-\sum_{j < k} \log \sigma(z_j, z_k)$ at the N -gon gives the mode-independent confining potential $C_1(\xi) = (N-1)(1+\xi^2)/(1-\xi)^2$.*
- (c) Full identification. *The constrained second variation (angular + radial at fixed angular impulse), divided by $2h$ per chiral sector, is*

$$\lambda_m = \underbrace{C_1(\xi)}_{\text{radial confinement}} - \underbrace{m(N-m)/2}_{\text{one-chiral OPE repulsion}}. \quad (5.3)$$

This is the Havelock eigenvalue, derived entirely from boundary data.

Proof. Part (a) is Proposition 5.1. For part (b): the log-radial second variation of the hyperbolic energy gives $C_1(\xi)$ as in (4.2) (Paper I, equation 5.7) (the Riemannian Havelock identity on $\mathbf{H}^2$). Part (c) combines (a) and (b) via the standard Lagrange-multiplier argument (Paper I, §3).

Identity block dominance for $N > 4$. The theorem requires that the identity Virasoro block dominates the OPE decomposition in each $\mathbb{Z}_N$ channel. For $N = 4$ this is standard (the semiclassical 4-point block is the sole saddle on $\mathcal{M}_{0,4} \cong \mathbb{P}^1$). For $N > 4$: the $\mathbb{Z}_N$ -symmetric N -point correlator of identical operators on $\mathbb{P}^1$ (*genus zero*) equals the 1-point function of the twist field σ_m on the orbifold $\mathbb{C}/\mathbb{Z}_N$ [?]. The orbifold has twisted-sector vacua $|k\rangle$ with conformal dimensions $h_k = k(N-k)/(2N^2) \times c > 0$ for $k = 1, \dots, N-1$. On genus zero, these contribute to the 1-point function only as virtual intermediate states, suppressed by $O(e^{-2\pi h_k}) = O(e^{-c})$ (exponentially small at large c , since $h_k \propto c$). The identity block gives the leading (classical) contribution $\exp(-c \times S_{\text{conical}})$; the $\mathbb{Z}_N$ symmetry reduces the moduli space $\mathcal{M}_{0,N+1}^{\mathbb{Z}_N}$ to dimension 2 (ring radius and overall phase). On this 2D space, the identity block dominates by two independent bounds:

1. *Eigenvalue bound (Proposition 7.1):* the $O(1/c)$ Virasoro corrections satisfy $|\delta f_m / \lambda_m| < 1/4$ for all non-critical m , so no eigenvalue changes sign at finite c and the symmetric saddle remains non-degenerate.

2. *OPE coefficient bound (Liouville action)*: the $\mathbb{Z}_N$ twist fields of order k have OPE coefficients suppressed by the Liouville action of the conical metric with opening angle $2\pi/N$ [??]: $C_k^2 \sim \exp(-b(N) \ln N)$. Since $b(N) \sim N^2/12$, this gives $C_k^2 \lesssim \exp(-N^2 \ln N/12)$ —exponential in N^2 suppression. The total non-identity contribution is bounded by $(N-1) \exp(-b(N) \ln N)$: $\sim 11\%$ at $N = 5$, $\sim 1.7\%$ at $N = 6$, $< 0.14\%$ for $N \geq 7$, and $< 10^{-10}$ for $N \geq 11$.

Bound (1) excludes competing saddles at the eigenvalue level; bound (2) controls the non-perturbative twist contributions to the conformal block decomposition. Together they establish identity block dominance for $N \geq 6$ (corrections $\leq 2\%$). At $N = 5$, the 11% correction is the leading systematic uncertainty. $\square$

Theorem 5.2 identifies λ_m as the m -th $\mathbb{Z}_N$ -channel stiffness of the semiclassical N -point function. No bulk computation is needed: the stability condition $\lambda_m > 0$ is *the confining potential exceeding the holomorphic OPE repulsion in channel m* , a statement in the boundary CFT. The palindromic thresholds $\xi^*(N)$ —at which $C_1(\xi^*) = f_{\max}(N)$ —are visible as algebraic-number bulk positions where the $\mathbb{Z}_N$ conformal block changes sign.

Quantum correction. The $\mathbb{Z}_N$ Fourier decomposition of the Virasoro block at central charge c confirms the identification at all orders. The m -dependent part of $\log F^{(m)}$ is

$$\left[\frac{m(N-m)}{2} - \frac{m^2(N-m)^2}{4c} + O(1/c^2) \right] \log q. \quad (5.4)$$

At $c = \infty$: the leading term is the Havelock Casimir (exact, verified for all $N = 4, \dots, 12$ and all modes). At finite c : the correction $\delta h_m = -f_m^2/c$ is a *polynomial in the Casimir*, so the algebraic structure (palindromic symmetry, number field) is preserved at each order in $1/c$. The fractional correction $|f_m/c| \leq 1/8$ (maximised at $m = N/2$, $c = N^2$) does not change any eigenvalue sign (Proposition 7.1).

6 The polygon–BTZ phase transition

The Euclidean action comparison between the polygon and the BTZ black hole makes the “Hawking–Page” language precise.

The Euclidean actions. At inverse temperature β , the regularised Euclidean actions of the competing configurations are:

$$I_{\text{polygon}}(\beta) = -\frac{\beta}{8G} + \beta E_{\text{polygon}}, \quad (6.1)$$

$$I_{\text{BTZ}}(\beta) = -\frac{\pi^2 \ell^2}{2G\beta} \quad (M = 0 \text{ threshold BTZ}), \quad (6.2)$$

where $E_{\text{polygon}} = Nm(1-2Gm) + V_{\text{int}}$ is the total energy of the N -gon (kinematic mass + gravitational binding), and $V_{\text{int}} < 0$ is the pairwise Havelock interaction on $\mathbf{H}^2$. The vacuum action $-\beta/(8G)$ is the standard thermal-AdS₃ contribution.

The transition temperature. Setting $I_{\text{polygon}} = I_{\text{BTZ}}$:

$$T_c(N, m, \xi) = \frac{1}{2\pi\ell} \sqrt{1 - 8G E_{\text{polygon}}(N, m, \xi)}, \quad (6.3)$$

which reduces to the standard Hawking–Page temperature $T_{\text{HP}} = 1/(2\pi\ell)$ for $E = 0$. For $T < T_c$: the polygon has lower action ($\Delta I < 0$, polygon preferred). For $T > T_c$: the BTZ has lower action. The total polygon energy E_{polygon} lowers T_c below T_{HP} : more mass in the polygon makes the BTZ transition harder, because the binding energy deepens the polygon potential well.

The negative mode. For $N > N_{\text{crit}}(\xi)$: the polygon has a negative Havelock eigenvalue $\lambda_m < 0$ at mode $m = \lfloor N/2 \rfloor$ (§12). In the Euclidean path integral, this is a *negative mode* of the gravitational action [Giombi et al., 2008]: a perturbation that lowers the action, signaling decay. The mode $m = \lfloor N/2 \rfloor$ displaces adjacent masses alternately inward and outward, a pattern that drives pairwise coalescence. In the nonlinear regime, the coalescing masses form a single body exceeding the BTZ threshold $8GM > 1$, completing the decay to a BTZ black hole.

The three-region phase diagram. The polygon–BTZ competition has three distinct regimes:

1. *Stable polygon, preferred* ($N \leq N_{\text{crit}}, T < T_c$): all Havelock eigenvalues are positive and the polygon has lower Euclidean action than the BTZ. The polygon contributes as a stable saddle to the path integral.
2. *Stable polygon, metastable* ($N \leq N_{\text{crit}}, T > T_c$): all eigenvalues are positive but the BTZ has lower action. The polygon is a local minimum separated from the BTZ by a barrier; the tunnelling rate is $\Gamma \sim e^{-S_{\text{BTZ}}}$.
3. *Unstable polygon, decay* ($N > N_{\text{crit}}$): the mode $m = \lfloor N/2 \rfloor$ is a negative mode, and the polygon decays continuously (no barrier) toward the BTZ. The palindromic threshold $\xi^*(N)$ marks the boundary between regions 1/2 and region 3.

The critical exponent. At the palindromic threshold $\rho = \rho^*(N)$, the critical mode $m^* = \lfloor N/2 \rfloor$ vanishes linearly: $\lambda_{m^*}(\rho) \sim \coth(\rho^*) \cdot (\rho - \rho^*)$. The one-loop partition function (Gaussian integration over the zero mode) gives the universal critical exponent $\nu = 1/2$:

$$Z(\rho) \sim \frac{Z_{\text{frozen}}}{|\rho - \rho^*|^{1/2}}, \quad Z_{\text{frozen}} = \frac{2^{3(N-2)/4}}{(N-3)!!}$$

where $(N-3)!! = 1 \cdot 3 \cdot 5 \cdots (N-3)$ is the double factorial (even N , one zero mode at $m^* = N/2$; for odd N , the palindromic pair m^* and $N-m^*$ both have $\lambda = 0$, giving two zero modes and $Z \sim |\rho - \rho^*|^{-1}$. Z_{frozen} for odd N is $\prod_{m \neq m^*, m \neq N-m^*} [f(m^*, N) - f(m, N)]^{-1/2}$, computed from the Casimir gaps $f(m^*, N) - f(m, N) = [m^*(N-m^*) - m(N-m)]/2$. The closed-form expression includes the normalisation of the first-order Kirchhoff path integral measure; it is verified numerically against the gap-by-gap product for all $N \leq 20$ (test suite: `test_gravity_partition.py`). The polygon–BTZ transition is second-order with mean-field exponents.

The entropy gap. In the microcanonical ensemble (fixed total energy), the BTZ *always* has higher entropy: $\Delta S = S_{\text{BTZ}}(E_{\text{polygon}}) > 0$. The polygon is therefore always entropically disfavored; its survival as a metastable state depends entirely on the positivity of the Havelock eigenvalues (the energy barrier), not on entropy. This makes N_{crit} the sharp dividing line: below it, the polygon is a long-lived metastable state; above it, the polygon decays on a dynamical timescale $\tau \sim 1/\sqrt{|\lambda_{[N/2]}|}$.

Microscopic entropy accounting. The BTZ entropy decomposes into discrete and continuous contributions:

$$S_{\text{BTZ}}(N) = \underbrace{S_{1\text{-loop}}(N)}_{\text{continuous}} + \underbrace{\Delta S_{\text{wc}}}_{\text{discrete}}, \quad (6.4)$$

where $S_{1\text{-loop}} = \log(N-3)!! - \frac{3(N-2)}{4} \log 2$ (from the Casimir gaps) and ΔS_{wc} is the wall-crossing contribution. At each palindromic threshold, the $\mathbb{Z}/2$ Galois action ($m \leftrightarrow N-m$) contributes:

N parity	Critical mode	ΔS
Even (self-paired $m = N/2$)	one eigenvalue crosses	$\log 2$ (1 bit)
Odd (paired $m, N-m$)	two eigenvalues cross	$2 \log 2$ (2 bits)

From $N = 7$ (the flat-plane marginal threshold) through $N = 12$: exactly 9 bits (verified numerically). Through $N = 17$ (11 wall-crossings): the discrete part contributes $17 \ln 2 \approx 11.8$ nats (9.9%), the continuous one-loop part 107.8 nats (90.1%). The L^2 -Morse index $\tau(P_-)$ has denominator 2 at each threshold (from the eta invariant shift of $1/2$ due to the zero mode), consistent with the K -theoretic framework.

The one-loop correction. The one-loop partition function around the polygon saddle is $Z_{1\text{-loop}} = (\det' \Delta_{\text{Lich}})^{-1/2}$, where Δ_{Lich} is the Lichnerowicz operator on the multi-cone geometry. By the conformal invariance of the flat-plane backreaction (Proposition 10.2), the leading correction comes from the spatial topology: the spectral bound $|\delta C_1| \leq 2.86/(4\pi\lambda_1)$ (where λ_1 is the spectral gap of the spatial Laplacian) gives a rigorous bound on the quantum shift of the stability eigenvalues. On the Bolza surface ($\lambda_1 \approx 3.84$): $|\delta C_1| < 0.06$, which does not shift N_{crit} away from 7 on the flat plane. On the modular surface ($\lambda_1 \approx 91$): $|\delta C_1| \lesssim 0.002$, negligible. The Bolza surface carries the $\{8, 3\}$ hyperbolic lattice on which ? constructed the hyperbolic Haldane model (Chern insulator with quantised Bott index $B = \pm 1$ and chiral edge states). The vortex stability analysis on the same $\{8, 3\}$ lattice shares the spectral data (λ_n of Δ_S) with the hyperbolic Haldane model. The bound $|\delta C_1| < 0.06$ shows that polygon stability is insensitive to the hyperbolic lattice topology.

Lorentzian continuation and the BTZ horizon. The curvature coefficient $C_1 = \log(2 \sinh \rho) + b(N)$ admits an analytic continuation $\rho \rightarrow \rho + i\pi/2$ that interchanges the BTZ exterior and interior: $\sinh(\rho + i\pi/2) = i \cosh \rho$, so $C_1^{\text{int}} = \log(2 \cosh \rho) + b(N) + i\pi/2$. The analytically continued kernel interpolates between curvature signs:

$\text{Im}(\rho)$	0	$\pi/2$	π
Curvature sign	$K=-1$	$K=0$	$K=+1$
Continued kernel	$\log(2 \sinh \rho)$	$\log(2 \cosh \rho) + i\pi/2$	$\log(2 \sinh \rho) + i\pi$

(The real parts of the continued kernels reproduce the pair interactions on spaces of the corresponding curvature sign; the imaginary parts are phases.) At each palindromic threshold $\rho^*(N)$: the critical eigenvalue λ_{m^*} crosses zero (*real* side), while the interior resonance has $\text{Im}(\lambda) = \pi/2$ (universal, independent of N and m). The polygon–BTZ transition is a *time–space exchange*: for $\rho > \rho^*$ the mode oscillates in time ($\omega = \sqrt{\lambda_{m^*}}$); for $\rho < \rho^*$ it grows spatially ($\kappa = \sqrt{|\lambda_{m^*}|}$); at $\rho = \rho^*$ the Killing vector degenerates.

Under Lorentzian continuation ($\rho \rightarrow \rho + i\varepsilon$), the partition function acquires a branch cut at the threshold: $Z_L \sim A(N) (\rho - \rho^* + i\varepsilon)^{-1/2}$ with spectral function $\text{Im}[Z_L] \propto |\rho - \rho^*|^{-1/2}$ (a van Hove singularity). The exponent $-1/2$ (divergent density at the edge) is the spectral signature of the integrable Havelock system. The Dyson circular ensemble (GUE, $\beta = 2$) has the opposite exponent: $\rho \propto |\lambda - \lambda_{\text{edge}}|^{+1/2}$ (vanishing density, the Wigner semicircle edge). The sign flip $-1/2 \rightarrow +1/2$ marks the transition from integrable (polygon) to chaotic (BTZ black hole) spectral statistics [?].

7 The Wheeler–DeWitt equation on the palindromic minisuperspace

The single degree of freedom ρ (geodesic radius of the polygon) and the potential $V(\rho) = \lambda_{m^*}(\rho)$ (the critical Havelock eigenvalue) define a minisuperspace Wheeler–DeWitt equation:

$$\left[-\frac{1}{2c} \frac{d^2}{d\rho^2} + \log(2 \sinh \rho) + b(N) - \frac{m(N-m)}{2} \right] \Psi = 0, \quad (7.1)$$

with $c = 12b(N)$ and $\hbar = 1$ (the semiclassical limit is $c \rightarrow \infty$). On constant-curvature $\mathbf{H}^2$ the Weyl anomaly $\delta_m = 0$ identically (§8), so it does not appear.

Notation: two forms of C_1 . The WDW potential uses the *Green’s function offset* $C_1^{\text{GF}}(\rho) = \log(2 \sinh \rho) + b(N)$, which diverges as $\rho \rightarrow 0$. The holographic sections use the *constrained angular stiffness* $C_1(\xi) = (N-1)(1+\xi^2)/(1-\xi)^2$ (eq. 4.2), which gives $C_1 \rightarrow N-1$ as $\xi \rightarrow 0$. These are *distinct* quantities: the constrained form subtracts the self-energy divergence via the angular-impulse constraint, leaving a finite limit. Both share the same large- ρ behaviour ($\sim \rho$ for $\rho \gg 1$) and the same $O(\varepsilon^2)$ curvature coefficient ($R/4$) in the small-ring expansion. The WDW potential is $V(\rho) = C_1^{\text{GF}}(\rho) - f(m^*, N)$. The WDW equation uses the unconstrained form because it retains the full ρ -dependence of the Green’s function.

Why second-order dynamics from a first-order system. The Kirchhoff equations are first-order, but the WDW equation for the breathing mode ρ is second-order. Two independent derivations combine:

1. *The kinetic term exists* (Caldeira–Leggett mechanism, Proposition 12.2): the $N-2$ angular modes form a harmonic bath whose exact influence functional (Gaussian, linear coupling) produces a $\dot{\rho}^2$ noise kernel in the Onsager–Machlup path integral.
2. *The coefficient is $c = 12b(N)$* (self-consistency, Proposition 8.2): $b(N)$ is the unique mode-independent offset in the eigenvalue decomposition, computed from the csc^2 kernel to 10^{-16} precision (Paper I). The CL noise kernel produces a *different* kinetic coefficient ($\sim N^2 \rho^{*2} / ((N-2) \coth^2 \rho^*)$), which does not equal $12b(N)$ at any N . The value $c = 12b(N)$ is instead the unique coefficient for which the WDW ground-state eigenvalue reproduces the classical Havelock spectrum—a *uniqueness condition*.

tion from the eigenvalue structure, not a dynamical derivation from the noise kernel. No parameter is fitted: $b(N)$ is determined by the kernel.

Each term has a definite origin:

Term	Expression	Origin
Quantum kinetic	$-\hbar^2/(2c) \Psi''$	Todd offset $c = 12 b(N)$ (§1)
Classical potential	$\log(2 \sinh \rho)$	$\mathbf{H}^2$ Green's function
Vacuum energy	$b(N) \sim c/12$	Orbifold CFT vacuum (Todd class)
Matter	$m(N-m)/2$	Twist-field dimension (first Chern class)
One-loop	δ_m	Weyl anomaly (traceless, palindromic)

Three regimes.

1. **Classical** ($c \rightarrow \infty$): the kinetic term vanishes; the constraint reduces to $V(\rho) = 0$, i.e., $\lambda_{m^*}(\rho^*) = 0$ —the palindromic threshold, which is the Einstein equation of the vortex-gravity system.
2. **Semiclassical** (WKB): $\Psi \sim \exp(\pm(i/\hbar) \int \sqrt{2cV} d\rho)$, oscillatory in the exterior ($V > 0$, sinh-branch) and evanescent in the interior ($V < 0$, cosh-branch). The continuation $\rho \rightarrow \rho + i\pi/2$ maps between them (§6, Lorentzian paragraph).
3. **Quantum** (near threshold): $\Psi \sim \text{Ai}((2c \coth \rho^*)^{1/3}(\rho - \rho^*))$, smoothing the classical horizon over the width

$$\delta\rho = \frac{1}{(2c \coth \rho^*)^{1/3}}, \quad (7.2)$$

where $\coth \rho^* = V'(\rho^*)$ is the surface gravity. For large ρ^* ($\coth \rho^* \rightarrow 1$): $\delta\rho \rightarrow (2c)^{-1/3} \approx (2N^2)^{-1/3}$. At $c = \infty$: $\delta\rho \rightarrow 0$ (sharp classical threshold). At finite c : the horizon has quantum width $O(c^{-1/3})$.

Proposition 7.1 (Eigenvalue non-crossing at finite c). *For $N \geq 4$ and $c \geq N^2$ (both satisfied for all $N \geq 4$; the bound requires $\lfloor N/2 \rfloor \geq 2$ for the maximisation, i.e. $N \geq 4$; $c \geq N^2$ holds since $c = 12b(N) > N^2$ for all $N \geq 3$ since $c = 12b(N) = N^2 + N - 12\ln 2 + O(\ln N/N) > N^2$ for all $N \geq 3$), the mode-dependent part of the $O(1/c)$ Virasoro block correction satisfies*

$$\frac{|\delta f_m - \delta f_{m+1}|}{f(m+1, N) - f(m, N)} = \frac{2m(N-m-1) + (N-1)}{2c} \leq \frac{1}{4} - \frac{1}{2N^2} < \frac{1}{4}. \quad (7.3)$$

In particular, no eigenvalue crossing occurs and the critical mode index $m_{\text{crit}} = \lfloor N/2 \rfloor$ is preserved.

Proof. The leading $1/c$ correction to the Casimir is $\delta f_m = -f_m^2/c = -m^2(N-m)^2/(4c)$. The difference factors as $\delta f_m - \delta f_{m+1} = (N-2m-1)[2m(N-m-1) + (N-1)]/(4c)$. Dividing by the Casimir gap $f(m+1) - f(m) = (N-2m-1)/2$ gives the ratio $R(m) = [2m(N-m-1) + (N-1)]/(2c)$. The maximum over $1 \leq m < N/2$ is at $m = N/2 - 1$, giving $R_{\text{max}} = (N^2-2)/(4N^2) = 1/4 - 1/(2N^2) < 1/4$. Since $R < 1$ for all adjacent pairs: the residual gap is at least $3(N-2m-1)/8 > 0$, and no reordering occurs. $\square$

Remark 7.1 (Algebraic field preservation). *The Zamolodchikov c -recursion gives the correction as a rational function of c , with only $\mathbb{Z}_N$ -invariant channels contributing. At $c = N^2$: the palindromic coefficients $A(c), B(c)$ remain in $\mathbb{Q}(\cos(2\pi/N))$ —the same number field as at $c = \infty$. The palindromic form $A\xi^2 + B\xi + A = 0$ is exact at all c (the conformal inversion $\xi \leftrightarrow 1/\xi$ is a geometric symmetry of $\mathbf{H}^2$, not a perturbative artefact).*

The palindromic threshold $\xi^*(N)$ is the critical AdS radius at which the polygon–BTZ transition occurs, and the algebraic number field $\mathbb{Q}(\sqrt{D})$ of the threshold is the arithmetic invariant of the phase boundary.

8 The three-layer decomposition

Proposition 8.1 (Orbifold–Havelock correspondence). *At leading order in $1/N$, the Havelock eigenvalue on $\mathbf{H}^2$ matches the $\mathbb{Z}_N$ orbifold CFT ground-state energy at $c_{\text{orb}} = 12N^2$ (the UV orbifold central charge) with angular impulse constraint:*

$$\lambda_m = -E_0(m; c_{\text{orb}}=12N^2) + \mu_L(\xi), \quad (8.1)$$

where $E_0(m) = -N/2 + m(N-m)/2$ is the twisted-sector ground state and $\mu_L(\xi) = C_1(\xi) - N/2$ is the Lagrange multiplier. The matching uses $c_{\text{orb}} = 12N^2$ (so that $c/(24N^2) = 1/2$), while the exact Havelock central charge is $c = 12b(N) = N(N+1) - 12\ln 2 + \dots \approx N^2$ at leading order. The distinction: c_{orb} is the UV (free-field) central charge of the $\mathbb{Z}_N$ orbifold; $c = 12b(N)$ is the exact IR value after regularisation of the vortex self-energy. The mode-dependent matching (the $m(N-m)/2$ term) uses only $c_{\text{orb}}/(24N^2) = 1/2$; the mode-independent $O(N)$ corrections (the $\ln 2$ and $(\ln N)/(N-1)$ terms in $b(N)$) enter through μ_L , not through E_0 .

Symbol	Value	Role
$c_{\text{orb}} = 12N^2$	UV orbifold	Mode-dependent matching
$c = 12b(N) \approx N^2 + N$	Exact (IR)	WDW kinetic coefficient
$c \approx N^2$	Leading order	Approximation in estimates

Proof. The orbifold ground-state energy $E_0(m) = -c/(24N) + cm(N-m)/(24N^2)$ at $c = 12b(N) \approx N^2$ gives $E_0 = -N/2 + m(N-m)/2$. Then $-E_0(m) + \mu_L = [N/2 - m(N-m)/2] + [C_1(\xi) - N/2] = C_1(\xi) - m(N-m)/2 = \lambda_m$. $\square$

Proposition 8.2 (Todd class decomposition of the eigenvalue offset). *The $\mathbb{Z}_N$ -equivariant eigenvalue decomposition of the vortex Laplacian on the cusped Poincaré disk (analogous to the equivariant Riemann–Roch theorem) satisfies*

$$\text{ind}_m = \underbrace{\frac{m(N-m)}{2}}_{c_1\text{-term}} + \underbrace{b(N)}_{\text{Todd term}} + \delta_m, \quad (8.2)$$

where $b(N) = N(N+1)/12 - \log 2 + \log N/(N-1)$ is independent of both m and ρ (§1, verified in Paper I). The central charge $c = 12b(N) = N^2 + O(N)$ is the same for every mode m , extracted from the Todd correction alone.

Proof. The Havelock eigenvalue satisfies $\lambda_m + f_m = C_1(\rho) + \delta_m$, and $C_1(\rho) = \log(2 \sinh \rho) + b(N) + O(e^{-\rho})$ (the radial dependence is absorbed by the Green’s function). Subtracting

$\log(2 \sinh \rho)$: $\text{ind}_m := \lambda_m + f_m - \log(2 \sinh \rho) = b(N) + \delta_m$. Since δ_m is traceless ($\sum \delta_m = 0$ to 10^{-15}) and palindromic ($\delta_m = \delta_{N-m}$), the mode-independent part $b(N)$ is extracted exactly. $\square$

The three-layer decomposition $\lambda_m = C_1 - f_m + \delta_m$ is the *spectral equivariant Riemann–Roch theorem*: f_m is the c_1 -contribution (twist-field dimension), $b(N)$ is the Todd-class correction (vacuum energy $\approx c/12$), and δ_m is the one-loop remainder (the Weyl anomaly). The central charge is not a fitted parameter: it is $c = 12 b(N)$, determined by the index theorem.

Exact Weyl anomaly formula. *Clarification:* The anomaly δ_m below is the *raw* three-layer residual for the *unconstrained* eigenvalue on the flat plane. On *constant-curvature* surfaces with the *constrained* (angular-impulse) eigenvalue, $\delta_m = 0$ identically (Paper IV, Theorem 7.1). The nonzero formula below applies to the flat-plane or non-constant-curvature case.

The anomaly δ_m decomposes exactly into character sums on $\mathbb{Z}/N\mathbb{Z}$:

$$\delta_m = L(m, N) - \frac{1}{4} S_2(m, N) + c(N), \quad (8.3)$$

where $L(m, N) = \sum_{p=1}^{N-1} \cot(\pi p/N) \cos(2\pi p m/N)$ is the cotangent sum, $S_2(m, N) = \sum_{p=1}^{N-1} \csc^2(\pi p/N) \cos(2\pi p m/N)$ is the cosecant-squared sum, and $c(N)$ is the unique constant enforcing $\sum_{m=1}^{N-1} \delta_m = 0$ (tracelessness). The coefficients $a = 1$, $b = -1/4$ are exact and universal ($R^2 = 1.000000$ for all tested $N = 4, \dots, 24$). This identifies the anomaly as the lattice correction (orbifold fixed-point contribution in the KK picture) to the continuum Casimir.

The value $c = 12 b(N) \approx N^2$ makes five independently testable predictions beyond the Casimir: (i) the exact offset $b(N) = N(N+1)/12 - \log 2 + \log N/(N-1)$ (Paper I); (ii) the mean aliasing $\langle D \rangle = N(N+1)/12$ (exact); (iii) the growth law $\rho^*/f_{\text{crit}} \rightarrow 1/3$ (Paper I); (iv) the quartic coupling $Q/f_{\text{crit}}^2 = N/48$ (exact for even N); (v) the mode-independent Laplacian $\Delta_{\mathbf{H}^2} h = -1/2$. All five follow from $c = 12 b(N) \approx N^2$ through the B_2 -tower: the Casimir $m(N-m)/2$ is $O(c)$; the Weyl anomaly δ_m is $O(c^0)$ (topological); the Virasoro block correction is $O(1/c) \approx 10^{-5}$ (§4).

The tower extends to a *Magri bihamiltonian hierarchy*. The Thomson vortex system IS the Calogero–Moser–Sutherland model at coupling $g = 1$ (the Havelock interaction matrix $M_{jk} = 1/\sin^2(\pi(j-k)/N)$ is exactly the CMS Hamiltonian; see ?). The CMS model is bihamiltonian [?]: the Kirchhoff bracket P_0 (mode-independent) and the Casimir bracket P_1 (eigenvalue f_m on mode m) are compatible, with recursion operator $R = P_1 P_0^{-1}$ acting as multiplication by $f_m = m(N-m)/2$. By the Magri theorem, the k -th conserved quantity I_k has mode eigenvalue f_m^k , with physical derivatives $d_{2k} = \alpha_k N f_m^k$ ($\alpha_1 = 1$, $\alpha_2 = 4$, $\alpha_3 = 64$, verified numerically for $N \leq 16$). The quartic coupling $Q = (N/48)f^2$ lies inside the hierarchy ($H_4^{\text{diag}} \propto I_2$), guaranteeing $\{I_j, H_4\} = 0$ by integrability. The three-layer decomposition $\lambda_m = C_1 - f_m + \delta_m$ is the truncation of this hierarchy at level 1.

On arithmetic surfaces (Bolza, modular, etc.), the eigenvalue acquires a topological correction that is invisible to the confining potential $C_1(\xi)$:

$$\lambda_m = \underbrace{C_1(\xi)}_{\text{Ricci scalar}} - \underbrace{\frac{m(N-m)}{2}}_{\text{Casimir}} + \underbrace{\delta_m(N)}_{\text{Weyl anomaly}}. \quad (8.4)$$

Each layer has a distinct character:

1. $C_1(\xi)$: continuous, surface-dependent, mode-independent. Scales approximately as $C_1 \propto \rho$ over the range $\rho \in [0.01, 10]$ (with $R^2 = 0.98$, where ρ is the geodesic radius and $\xi = \tanh^2(\rho/2) \in (0, 1)$); the exact formula is eq. (4.2).
2. $f(m, N) = m(N-m)/2$: discrete, universal (surface-independent). The $\mathbb{Z}_N$ Casimir from the Havelock identity.
3. $\delta_m(N)$: topological, UV-finite, ξ -independent (variation $< 10^{-13}$), and exactly traceless ($\sum_{m=1}^{N-1} \delta_m = 0$ to 10^{-15}). Carries all number-theoretic content: Hecke eigenvalues, palindromic structure, Langlands data.

By structural analogy with the decomposition of the Riemann curvature tensor: C_1 plays the role of the Ricci scalar, $f(m, N)$ the traceful part, and δ_m the traceless Weyl-like anomaly. In $\text{AdS}_3/\text{CFT}_2$, the Brown–Henneaux scaling $C_1 \propto \log(\xi)$ says that the Weyl-like anomaly δ_m is the topological correction that carries the arithmetic of the underlying surface, contributing corrections of order $\|\delta\|/C_1 \sim 2\%$ at typical parameter values.

The Weyl norm scales as $\|\delta\|^2 \sim N^{4.2}(\log N)^{4.1}$ for $N \geq 40$ (not a pure power law). The triangle ($N = 3$) is *anomaly-free*: $\delta_m = 0$ exactly for all Green’s functions and curvatures, because $\cos(2\pi/3) = \cos(4\pi/3)$ forces $G_1 = G_2$. Under Casimir-conserving mergers $N_1 + N_2 \rightarrow N_3$ ($N_3^3 - N_3 = N_1^3 - N_1 + N_2^3 - N_2$): the unique anomaly-*destructive* reaction is $4+4 \rightarrow 5$ (88% Weyl loss); all other mergers create Weyl content (gains 46% to 849%). The Weyl entropy per degree of freedom $s(N) = \log \|\delta\|^2 / \sum f$ peaks at $N \approx 9$ –10, coinciding with the observed planetary polygon range ($N = 5, 6, 8$).

The anomaly δ_m has two independent origins, which the parameter space (Δ, ξ) separates cleanly:

1. *Interaction anomaly* ($\Delta > 0$ on $\mathbb{R}^2$): deviation of $V = r^{-2\Delta}$ from $-\log r$. Large ($\|\delta\|^2 = O(1\text{--}100)$) but arithmetically trivial.
2. *Geometric anomaly* ($\Delta = 0$ on $\mathbf{H}^2$): deviation of the hyperbolic Green’s function from the flat one. Small ($\|\delta\|^2 = O(0.01\text{--}100)$) but carries all number theory: palindromic polynomials, Hecke eigenvalues, Langlands data.

The RG flow ($\Delta \rightarrow 0$) strips away the large, trivial interaction anomaly, leaving behind the small geometric anomaly as the residue that survives at the conformal fixed point. The number theory is not emergent from the flow; it is a property of the conformal fixed point, activated by curvature. The three-layer structure has a symplectic origin: the Kirillov form ω_{KR} (mode-independent) and the Weil–Peterson form ω_{WP} (mode-dependent) are *not* proportional; their discrepancy $\omega_{\text{WP}} - \alpha \omega_{\text{KR}} \propto f(m, N)$ is the Casimir (Layer 2), and the residual beyond this linear relation is the Weyl anomaly (Layer 3).

The B_2 organising principle — connecting the stability polynomial $6B_2(x) = P_2(2x-1)$ to Gauss–Legendre quadrature, the Euler–Maclaurin duality, and the csc^2 kernel universality (Calogero–Moser–Sutherland, Dyson, Haldane–Shastry, Laughlin, Selberg, W_N , AGT/Nekrasov) — is developed in Paper I, §??.

9 Higher-dimensional extension: the $\text{CFT}_3/\text{AdS}_4$ Casimir

The two-dimensional results extend to conformal field theory in any dimension d via the power-law interaction $V(x, y) = |x - y|^{-2\Delta}$, where Δ is the conformal dimension. For $\Delta \rightarrow 0$: $V \rightarrow -\log |x - y|$ (the 2D case). For $\Delta = 1$: the CFT_3 two-point function on S^2 (the boundary of AdS_4).

Proposition 9.1 (Generalised Havelock Casimir). *For N equally-spaced points on a ring on S^2 , interacting via $V = |x - y|^{-2\Delta}$, the angular Hessian eigenvalues factorise as $\mu_m = g(\theta_0, \Delta) \cdot h(m, N, \Delta)$, where the universal Casimir is*

$$h(m, N, \Delta) = \sum_{p=1}^{N-1} \frac{1 + 2\Delta \cos^2(\pi p/N)}{\sin^{2\Delta+2}(\pi p/N)} (1 - \cos \frac{2\pi m p}{N}). \quad (9.1)$$

For $\Delta = 0$: $h(m, N, 0) = 2m(N-m)$. The ratio $h(m)/h(1)$ is independent of θ_0 , confirming that the Havelock decomposition extends to all Δ .

The constrained Hessian on the flat plane with $V = |x - y|^{-2\Delta}$ gives an N -dependent stability boundary $N_{\text{crit}}(\Delta)$:

Δ	0	0.5	1	2	5	10
N_{crit}	7	5	5	5	4	4

The classical $N_{\text{crit}} = 7$ is an isolated phenomenon of the logarithmic interaction. For the CFT₃ value $\Delta = 1$: $N_{\text{crit}} = 5$.

Under the Green's function identification $\Delta = (d-2)/2$, the table acquires a dimensional interpretation:

Dimension d	2	3	4	5	≥ 6
$\Delta = (d-2)/2$	0	1/2	1	3/2	≥ 2
N_{crit}	7	5	5	5	4-5

The hexagon ($N = 6$, Saturn) is a *uniquely two-dimensional* phenomenon: it is marginally stable at $d = 2$ ($\lambda_{\text{min}} = 1/2$) but unstable for $d \geq 3$ ($\lambda_{\text{min}} = -0.02$ at $d = 3$; `test_ncrit_delta.py`, $N_{\text{crit}}(\Delta=0.5) = 5 < 6$). The pentagon ($N = 5$) survives in all tested dimensions $d \leq 10$.

Proposition 9.2 (Soft Havelock decomposition in dimension d). *For the interaction $V = r^{-(d-2)}$ ($d \geq 3$) on the N -gon ring, the angular Casimir decomposes as*

$$\sigma_m^{(d)} = a(N, d) \frac{m(N-m)}{2} + R_m(N, d), \quad (9.2)$$

where $a(N, d)$ is the leading Casimir coefficient (determined by least-squares fit) and R_m is the residual. For all tested $N = 5, \dots, 20$ and $d = 3, 4, 5$: the residual satisfies $|R_m - R_{m'}| < a |f_m - f_{m'}|$ for every adjacent pair, so the mode ordering is preserved. The critical mode is always $m = \lfloor N/2 \rfloor$ (same as $d = 2$), and the stability threshold has the same palindromic form as the two-dimensional case, with coefficients that depend on d .

For $d = 2$ ($s = 0$): $a = 1$ exactly and $R_m = 0$ (the Havelock identity). This is the *unique* dimension where the Casimir is an exact polynomial—because $\csc^2(\pi p/N)$ has a closed discrete Fourier transform ($\sum \csc^2 \cdot (1 - \cos) = 2m(N-m)$), while $\csc^s$ for $s \neq 2$ does not. The *exact* algebraic structure of Paper I (palindromic polynomials, algebraic number fields, modular forms) is specific to $d = 2$, a consequence of the uniqueness of the $\csc^2$ weight function among power-law potentials. However, the *structural skeleton* survives in all dimensions:

1. **Mode ordering.** The critical mode is $m = \lfloor N/2 \rfloor$ in every d (Proposition 9.2). The residual R_m is too small to reorder eigenvalues.
2. **Palindromic symmetry.** The symmetry $\sigma_m = \sigma_{N-m}$ holds exactly for all d (from the discrete cosine: $\cos(2\pi pm/N) = \cos(2\pi p(N-m)/N)$). On curved spaces, conformal inversion $\xi \leftrightarrow 1/\xi$ is geometric (a property of the Poincaré ball, not of the interaction), so the threshold equation retains palindromic form.
3. **Finite- c algebraic structure.** The $O(1/c)$ Virasoro block correction $\delta h_m = -f_m^2/c$ is a *polynomial in the Casimir* (equation 5.4), preserving the algebraic structure at every order.
4. **Weyl anomaly shape.** The normalised anomaly profile $\delta_m/\|\delta\|$ has inner products > 0.98 between $\Delta = 0$ and $\Delta = 2$ for $N \leq 6$: the number-theoretic fingerprint transfers to $d = 3$ with $< 2\%$ distortion.

The $d = 2$ case is where the algebra is *exact*; the $d \geq 3$ case is where it is *approximate but structurally preserved*. The mechanism by which the logarithmic structure survives in 3+1D is the soft Havelock decomposition: the Casimir $m(N-m)/2$ is the dominant term at every d , with the interaction-specific residual R_m as a controlled perturbation.

Remark 9.1 (Transcendence of the transition dimensions). *The critical value $\Delta^*(N)$ where N_{crit} drops below N is expected to be transcendental for $N \geq 5$. The threshold equation involves $\sin^{-2\Delta}(\pi p/N)$ for different p with $\mathbb{Q}$ -linearly independent logarithms. For $N = 5$: the threshold equation involves $\sin^{-2\Delta}(\pi/5)$, an algebraic base raised to the power Δ . If Δ^* is algebraic and irrational, the Gelfond–Schneider theorem makes the individual terms transcendental, and Baker’s theorem on linear forms in logarithms would be needed to show no cancellation occurs. A rigorous proof requires Baker-type estimates. This contrasts with the palindromic thresholds $\xi^*(N)$ on $\mathbb{H}^2$, which lie in explicit algebraic number fields.*

For $N = 5$, the pentagon identity $\sin(2\pi/5)/\sin(\pi/5) = \varphi$ (the golden ratio) gives the eigenvalue as $\lambda_2(\Delta) = P_0(\Delta) + P_1(\Delta) \varphi^{-(2\Delta+2)}$, where P_0, P_1 are algebraic functions of Δ . The transition occurs at $\varphi^{2\Delta^*+2} = P_1/(-P_0) \approx 25.8$, giving $\Delta^* \approx 2.378$.

Theorem 9.3 (RG monotonicity of the Casimir ratio). *For any $N \geq 4$ and $m_{\text{crit}} = \lfloor N/2 \rfloor$, the generalised Casimir ratio $R(\Delta) = h(m_{\text{crit}}, N, \Delta)/h(1, N, \Delta)$ is strictly increasing on $[0, \infty)$, with exact limits*

$$R(0) = \frac{m_{\text{crit}}(N-m_{\text{crit}})}{N-1}, \quad R(\infty) = \left[\frac{\sin(\pi m_{\text{crit}}/N)}{\sin(\pi/N)} \right]^2. \quad (9.3)$$

Proof. By the Chebyshev identity, the mode amplification factor $f(p) = (1 - \cos(2\pi m_{\text{crit}} p/N))/(1 - \cos(2\pi p/N))$ achieves its global maximum at $p = 1$ (and its palindromic twin $p = N-1$):

Even $N = 2k$: $\sin(\pi k p/N) = \sin(\pi p/2) \in \{0, \pm 1\}$, so $f(p) = 1/\sin^2(\pi p/N)$ for odd p and $f(p) = 0$ for even p . Since $\sin(\pi p/N) \geq \sin(\pi/N)$ for $p \geq 2$: $f(p) \leq f(1)$.

Odd $N = 2k+1$: The identity $\sin(k\pi p/N) = \sin(\pi p/2 - \pi p/(2N))$ (using $k/N = 1/2 - 1/(2N)$) gives explicit formulas: $f(p) = 1/(4\sin^2(\pi p/(2N)))$ for odd p , and $f(p) = 1/(4\cos^2(\pi p/(2N)))$ for even p . Since $f(1) = 1/(4\sin^2(\pi/(2N)))$: for odd $p \geq 3$, $\sin(\pi p/(2N)) \geq \sin(\pi/(2N))$ (monotonicity of $\sin$ on $(0, \pi/2)$), so $f(p) \leq f(1)$; for even

$p \leq N-1$, $\cos(\pi p/(2N)) \geq \sin(\pi/(2N))$ (since $\pi p/(2N) + \pi/(2N) \leq \pi/2$ iff $p \leq N-1$), so $f(p) \leq f(1)$.

The normalised weight $\mu_p(\Delta) \propto \tilde{w}(p, \Delta) c_1(p)$ concentrates on $\{1, N-1\}$ as $\Delta \rightarrow \infty$, because $\tilde{w}(p)/\tilde{w}(1) = O([\sin(\pi/N)/\sin(\pi p/N)]^{2\Delta}) \rightarrow 0$ for $p \notin \{1, N-1\}$. Since $R(\Delta) = \mathbb{E}_\mu[f]$ and μ concentrates at the global maximum of f : $dR/d\Delta = \text{Cov}_\mu(d \log \tilde{w}/d\Delta, f) > 0$ (both f and $g = d \log \tilde{w}/d\Delta = -2 \log \sin(\pi p/N)$ are maximised at $p = 1$ and monotonically ordered on $\{1, \dots, \lfloor N/2 \rfloor\}$; the Chebyshev sum inequality gives $\text{Cov}_\mu(g, f) > 0$). $\square$

Corollary 9.4 (Stability monotonicity). $N_{\text{crit}}(\Delta)$ is non-increasing in Δ : once a polygon type loses stability, it never regains it. The quantity $\log N_{\text{crit}}(\Delta)$ decreases from $\log 7$ ($\Delta = 0$, the Havelock regime) to $\log 3$ ($\Delta \rightarrow \infty$, only triangles survive). This is reminiscent of the Zamolodchikov c -theorem, though N_{crit} is a step function (not smooth) and Δ is not an RG parameter.

10 Triangle universality and dynamical triangulation

The equilateral triangle ($N = 3$) is the *unique* regular polygon that is dynamically stable for all interaction ranges and quantum deformation levels.

Theorem 10.1 (Triangle universal stability). *The equilateral triangle ($N = 3$) is the unique regular polygon that is stable on the flat plane for all $\Delta \geq 0$, all TQFT levels $k \geq 3$, and whose tessellation $\{3, 6\}$ is the unique stable regular lattice for all $\Delta > 0$.*

Proof sketch. Three independent arguments:

1. *Equal chords.* The triangle is the unique regular polygon where all chord distances are equal ($\sin(\pi/3) = \sin(2\pi/3) = \sqrt{3}/2$). No range-dependent competition between nearest and next-nearest neighbors $\implies$ stability is Δ -independent.
2. *Single mode.* The triangle has one nontrivial $\mathbb{Z}_3$ mode ($m = 1$, with $m = 2$ its palindromic conjugate). The Havelock eigenvalue $\lambda_1 = (N-1) - 1 = 1 > 0$. Mode competition requires at least two independent modes ($N \geq 4$).
3. *Zero Weyl anomaly.* The anomaly δ_m is traceless: $\sum \delta_m = 0$. With one mode: $\delta_1 = 0$ identically. Verified: anomaly norm = 2.2×10^{-16} (machine zero) on the square torus.

For the tessellation claim: the dynamical matrix (phonon spectrum) of the triangular lattice under $V = r^{-2\Delta}$ has all eigenvalues positive for *every* $\Delta > 0$ (verified by lattice sums with convergence study and Richardson extrapolation). The minimum phonon frequency is at the K -point of the Brillouin zone. The square lattice (coordination 4) has a permanent X -point instability (stripe mode: $\omega^2(X) < 0$ for all Δ); the hexagonal lattice (coordination 3) has a permanent K -point instability. The triangular lattice is the *only* regular lattice that is ever stable, at any interaction range. $\square$

Remark 10.1 (Dynamical triangulation). *In causal dynamical triangulations (CDT) and Regge calculus, spacetime is discretised into simplices (triangles in 2D). This triangulation is imposed by hand as a regularisation. Theorem 10.1 suggests it may be imposed by physics: among regular lattices, the triangular lattice is the unique stable configuration, so the gravitational path integral $Z = \sum_{\text{tilings}} e^{-S}$ may be dominated by triangulated configurations—the square and hexagonal alternatives are saddle points with negative phonon modes at all Δ . The extension to irregular triangulations (as used in CDT) remains open.*

11 The polygon partition function

The one-loop partition function of the vortex system organises naturally by algebraic number field.

Proposition 11.1 (Polygon partition function). *The partition function for the vortex-polygon system (summing over polygon number N , not over geometries) on $\mathbf{H}^2$ is*

$$Z_{\text{grav}} = \sum_{N=3}^{\infty} e^{-b(N)} Z_{\text{frozen}}(N), \quad (11.1)$$

where $b(N) = N(N+1)/12 - \log 2 + \log N/(N-1)$ is the gravitational action and $Z_{\text{frozen}} = 2^{3(N-2)/4}/(N-3)!!$ is the one-loop determinant (even N ; the odd- N case is computed from Casimir gaps). The series converges super-exponentially: $e^{-b(N)} \sim e^{-N^2/12}$. Numerically, $Z_{\text{grav}} = 1.3304$ and $N = 3, \dots, 8$ capture 99.99% of the total.

Proof. Convergence: $e^{-b(N)} = O(e^{-N^2/12})$ decays faster than any polynomial, while $Z_{\text{frozen}} = O(N^{O(N)})$ grows sub-exponentially; the product vanishes. The numerical value $Z = 1.3304$ is computed by direct summation to $N = 20$ (the tail $N > 20$ contributes $< 10^{-20}$). $\square$

The partition function decomposes by *trace field*—the algebraic number field generated by $\cos(2\pi/N)$:

Trace field	N values	Fraction of Z
$\mathbb{Q}$ (rational)	3, 4, 6	90.98%
$\mathbb{Q}(\sqrt{5})$ (golden)	5, 10	8.26%
$\mathbb{Q}(\sqrt{2})$ (silver)	8	0.42%
$\mathbb{Q}(\cos 2\pi/7)$ (cubic)	7	0.34%
All higher fields	≥ 9	$< 0.01\%$

The rational field dominates; the golden ratio field gives the leading correction; the entire algebraic tower is exponentially suppressed by $e^{-N^2/12}$.

The Baum-Connes assembly map $\mu: K_0^{\mathbb{Z}_N}(\mathbf{H}^2) \rightarrow K_0(C_r^*(\mathbb{Z}_N))$ sends the equivariant index $\text{ind}_m = f(m, N) + b(N)$ to the K-theory class with trace $c = 12b(N)$. This is mode-independent (the *same* c for all m) and approximately integer: $c = N^2 + O(N)$ for all tested $N = 3, \dots, 24$. The discrepancy $c - N^2 \approx N + O(1)$ is the boundary self-energy ($N/12$) plus regularisation terms.

12 From polygon stability to the Einstein equations

We now present the central result of the gravitational theory: the Havelock eigenvalue structure *characterises* the vacuum Einstein equations. In $d = 2$ (2+1D), the three hypotheses below are *derived* from the vortex dynamics—making this a genuine derivation of the Einstein equations from polygon stability. In $d \geq 3$ (3+1D and higher), hypothesis (H1) becomes *equivalent* to the vacuum Einstein equations (see Remark 12.2); the theorem then provides a new characterisation of Einstein manifolds via polygon stability, rather than a one-directional derivation. The full 4D Einstein equations are derived independently in Paper IV through the boundary Virasoro mechanism (the emergent spin-2 graviton of Theorem 4.1). We state the theorem first, then collect the proof.

Theorem 12.1 (Polygon entropy maximisation implies constant curvature). *Let (S, g) be a two-dimensional Riemannian manifold. Place N equal point vortices in a regular polygon ring of geodesic radius ε centred at $x \in S$, interacting via the scalar Green's function of (S, g) . If:*

- (H1) Havelock structure: *the angular Hessian eigenvalues have the form $\lambda_m = C_1(x, \varepsilon) - m(N-m)/2$ with C_1 independent of m . Proved in $d = 2$ by the csc^2 uniqueness theorem (Paper I).*
- (H2) Entropy equilibrium: *the frozen-mode entropy $S(x) = \frac{1}{2} \sum_{m \neq m^*} \log \lambda_m(x, \varepsilon)$ is maximised over the surface — i.e., the polygon system is in thermodynamic equilibrium. Derived from the Onsager selection principle (Paper I, §5): among all configurations consistent with fixed total energy, the maximum-entropy state has $C_1(x, \varepsilon_0) = f(m^*, N)$ uniformly in x (the constrained optimum distributes curvature equally at every point).*
- (H3) Hadamard universality: *condition (H1) holds at every point $x \in S$. Automatic: the Hadamard parametrix $G(x, y) = -(1/2\pi) \log d(x, y) + \gamma_S(x) + O(d^2)$ is universal on any Riemannian surface.*

Then the scalar curvature of S is constant: $R(x) = R_0$ for all x . In $2+1D$, this gives the vacuum Einstein equations $R_{\mu\nu} = \Lambda g_{\mu\nu}$ (Schur's lemma; the Weyl tensor vanishes identically in three dimensions).

The cosmological constant Λ is determined by a separate dynamical argument (the WDW equation, Proposition 12.2 below), not by the theorem above. The theorem establishes the *form* of the Einstein equations; the WDW equation determines their *content* (Λ).

Proof. The proof has four steps: (1) force the interaction to be logarithmic, (2) extract $R(x)$ from the Hadamard parametrix, (3) conclude $R = \text{const}$ from (H2) + Step 2, (4) promote to Einstein in any spacetime dimension.

Step 1. Polynomial Casimir forces logarithmic interaction. Condition (H1) requires the angular Casimir to be the exact polynomial $f(m, N) = m(N-m)/2$. By the csc^2 uniqueness theorem (Paper I, Lemma 4.1), this polynomial arises *only* from the weight function $\text{csc}^2(\pi p/N)$, which is the discrete Fourier transform of the logarithmic kernel $-\log|z - w|$. Therefore the interaction potential is $V(z, w) = -G_S(z, w)$, where G_S is the Green's function of the scalar Laplacian on (S, g) .

Step 2. Small-ring expansion reveals local curvature. The Green's function on any Riemannian surface has the short-distance expansion $G_S(z, w) = -(1/2\pi) \log d(z, w) + \gamma_S(z) + O(d^2)$, where $d(z, w)$ is the geodesic distance and γ_S encodes the global geometry. Summing over the N -ring at radius ε centred at x :

$$C_1(x, \varepsilon) = (N-1) \left[1 - \frac{R(x) \varepsilon^2}{4} + O(\varepsilon^4) \right], \quad (12.1)$$

where $R(x)$ is the scalar curvature at x . The mode-independence of the $O(\varepsilon^2)$ correction is *automatic*: C_1 is mode-independent by the Riemannian Havelock identity (Paper I, Theorem 4.2), so its small- ε expansion inherits this property at every order. The coefficient $1/4$ is verified directly from the exact $\mathbf{H}^2$ formula: expanding $C_1/(N-1) = (1+\xi^2)/(1-\xi)^2$ for small $\xi = \tanh^2(\varepsilon/2) \approx \varepsilon^2/4$ gives $C_1/(N-1) \approx 1 + \varepsilon^2/2 = 1 - R\varepsilon^2/4$ (since

$R = -2$ on unit $\mathbf{H}^2$). The Hadamard parametrix on a 2D surface gives $G(x, y) = -(1/2\pi) \log d - (R/12\pi)d^2 + O(d^4)$ (Garabedian 1964, §9.3). Summing the $O(d^2)$ term over the N -ring with $\sum_p \sin^2(\pi p/N) = N/2$ gives the $1/6$ contribution; the $1/12$ arises from the curvature correction to chord distances ($d_{\text{chord}} \neq d_{\text{geodesic}}$ at $O(\varepsilon^3)$). Together: $1/6 + 1/12 = 1/4$.

Step 3. (H2) + Step 2 force $R = \text{const.}$ By (H2), entropy maximisation forces $C_1(x, \varepsilon)$ to be x -independent (the uniform- C_1 configuration maximises S_{total} by Jensen's inequality and the concavity of S ; see below). Write C_1^{eq} for this uniform value. Substituting the expansion (12.1):

$$(N-1) \left[1 - \frac{R(x)\varepsilon^2}{4} + O(\varepsilon^4) \right] = C_1^{\text{eq}} \quad \text{for all } x.$$

The right side is x -independent. Therefore $R(x) = R_0$ for all x . The *value* of C_1^{eq} (and hence R_0 and Λ) is determined separately by the energy constraint: the WDW equation (Proposition 12.2) gives $C_1^{\text{eq}} = f(m^*, N)$ at the BO threshold, with $R_0 = -4[f(m^*, N) - (N-1)]/[(N-1)\varepsilon_0^2]$.

Why (H2) is essential — and why it is not circular. Conditions (H1) and (H3) alone are automatic on *any* Riemannian surface (the Hadamard parametrix gives a mode-independent $C_1(x, \varepsilon)$ regardless of whether $R(x)$ is constant). Without (H2), (H1) + (H3) carry no curvature content.

Hypothesis (H2) provides the dynamical input: on a surface with non-constant curvature, the frozen-mode entropy $S(x) = \frac{1}{2} \sum_{m \neq m^*} \log \lambda_m(x)$ varies from point to point (points with larger $|R|$ have smaller λ_m and lower entropy). The entropy $S(x)$ is strictly concave in C_1 : $d^2 S / dC_1^2 = -\frac{1}{2} \sum_{m \neq m^*} 1/\lambda_m^2 < 0$. The total entropy $S_{\text{total}} = \int_S S(x) dA$ is therefore maximised (Jensen's inequality) when $C_1(x)$ is uniform—i.e. when $R(x) = \text{const.}$ The constraint is fixed total polygon energy $E_{\text{total}} = \int_S V(\rho(x)) dA$ (the BO potential integrated over the surface, which depends on the metric through $\rho(x)$ and is *not* topologically fixed—unlike $\int R dA$, which is determined by the Euler characteristic). The Lagrange multiplier for this energy constraint is Λ .

The derivation of (H2) is therefore *not* circular: it does not assume constant curvature. It uses the second law (entropy maximisation of the polygon system, the Onsager selection principle of Paper I) on a *general* background, and the maximum-entropy configuration turns out to be constant curvature. The Einstein equations are the unique equilibrium of the polygon entropy.

The explicit $\delta S / \delta g$ computation. In 2D conformal gauge $g = e^{2\varphi} g_0$, the scalar curvature satisfies $\delta R / \delta \varphi = -2R$ and the area element $\delta(dA) / \delta \varphi = 2dA$. Using the chain $dS / dC_1 = (1/2) \sum 1/\lambda_m$ (the entropy gradient) and $dC_1 / dR = -(N-1)\varepsilon^2/4$ (the Hadamard expansion, Step 2), the functional derivative of S_{total} is:

$$\frac{\delta S_{\text{total}}}{\delta \varphi(x)} = \frac{(N-1)\varepsilon^2}{4} R(x) \sum \frac{1}{\lambda_m(x)} + 2S(x).$$

At the entropy maximum, this is constant for all x : the Euler–Lagrange equation is $F(R(x)) = \Lambda_\varphi$ where $F(R) = [(N-1)\varepsilon^2/4] R \sum (1/\lambda_m) + 2S$. The function F is strictly monotone in R (since $dS/dC_1 > 0$ and $dC_1/dR < 0$, so $dF/dR \neq 0$). A monotone function equal to a constant forces $R(x) = R_0 = \text{const.}$ No Jensen inequality is needed—the Euler–Lagrange equation is solvable and its unique solution is constant curvature.

Step 4. Mode-independence + $R = \text{const}$ implies Einstein in any spacetime dimension d_{st} . In $d_{\text{st}} = 3$ (2+1D): the Weyl tensor has $d_{\text{st}}(d_{\text{st}}+1)(d_{\text{st}}+2)(d_{\text{st}}-3)/12 = 0$ components, so $R = \text{const}$ alone gives $R_{\mu\nu} = \Lambda g_{\mu\nu}$.

In $d_{\text{st}} \geq 4$: $R = \text{const}$ gives only the *trace* of the Einstein equations. However, condition (H1) provides the missing information. The Hadamard parametrix in d dimensions gives $G(x, y) = U(x, y)/\sigma^{(d-2)/2} + V \log \sigma + W$, where $\sigma = d(x, y)^2/2$. The Weyl tensor W_{ijkl} enters the angular structure of V at order σ through the traceless Riemann component, producing an $\ell = 2$ (quadrupole) angular dependence. In 3+1D spacetime ($d_{\text{st}} = 4$), the Weyl tensor has $d_{\text{st}}(d_{\text{st}}+1)(d_{\text{st}}+2)(d_{\text{st}}-3)/12 = 10$ independent components; the $\ell = 2$ sector contributes $2\ell + 1 = 5$ of these (the traceless symmetric part). The $\mathbb{Z}_N$ Fourier modes $m = 1, \dots, N-1$ project onto specific angular harmonics. Due to the palindromic symmetry $\cos(2\pi pm/N) = \cos(2\pi(N-p)m/N)$, the number of *independent real* angular probes is $\lfloor N/2 \rfloor$, not $N-1$. To constrain all 5 quadrupole components of the $\ell = 2$ Weyl sector in spatial dimension $d = 3$ ($d_{\text{st}} = 4$), we need $\lfloor N/2 \rfloor \geq 5$, i.e. $N \geq 10$ (even) or $N \geq 11$ (odd). Verified numerically: the 5×5 cosine matrix $M_{mp} = \cos(2\pi pm/N)$ has rank 5 for $N = 10$ and $N = 11$, but rank 3 for $N = 6, 7$ (insufficient). Any nonzero Weyl tensor therefore makes C_1 depend on m for $N \geq 10$, violating (H1). Hence (H1) at every x with $N \geq 10$ forces the traceless Ricci tensor $R_{ij} - (R/d)g_{ij} = 0$ on each spatial slice (in $d = 3$ the spatial Weyl tensor vanishes identically; the polygon probes the 5 independent traceless Ricci components). The promotion to the full spacetime Einstein equations $R_{\mu\nu} = \Lambda g_{\mu\nu}$ (including time-space components) requires additionally the boundary Virasoro mechanism of Paper IV, which supplies the dynamical content in the time direction.

In $d_{\text{st}} = 5$ (4+1D), additional quadrupole components appear, requiring $N \geq 12$ (Paper IV, Theorem 15.3). $\square$

Proposition 12.2 (The cosmological constant from the WDW equation). *The value of Λ in Theorem 12.1 is determined by the Wheeler–DeWitt equation for the breathing mode ρ :*

$$\left[-\frac{1}{2c} \frac{d^2}{d\rho^2} + V(\rho) \right] \Psi = 0, \quad V(\rho) = C_1(\rho) - f(m^*, N).$$

Two independent ingredients determine this equation:

The mechanism (*why $\dot{\rho}^2$ appears*). The Kirchhoff equations are first-order; the breathing mode ρ is a collective coordinate with no canonical momentum. The $N-2$ angular modes form a thermal bath whose fluctuations drive ρ stochastically (Caldeira–Leggett mechanism). The Onsager–Machlup noise kernel gives $S_{\text{OM}} = \int [c/2 \dot{\rho}^2 + V] d\tau$. The CL conditions hold: (i) the angular modes are harmonic (exact Fourier diagonalisation); (ii) the coupling is mode-independent ($C'_1 = \coth \rho$ for all m); (iii) timescale separation ($\tau_{\text{tunnel}} \sim e^{S_{\text{BO}}} \gg 1$). The bath has $N-2$ discrete modes; the CL derivation is exact for any number of harmonic oscillators with linear coupling (Caldeira & Leggett 1983, eq. 3.1), not only in the continuum limit.

The coefficient (*why $c = 12b(N)$*). The coefficient c is not the CL friction $\gamma = \coth^2 \rho \Sigma(1/\lambda_m) \sim O(N)$; it is the algebraic invariant of the eigenvalue decomposition. The frozen-mode entropy is $S_{\text{bath}}(\rho) = \frac{1}{2} \sum_{m \neq m^*} \log \lambda_m(\rho)$, and $b(N)$ is the mode-independent offset in the eigenvalue decomposition $\lambda_m = C_1 - f_m + \delta_m$ with $C_1 = \log(2 \sinh \rho) + b(N)$ (Proposition 8.2). The coefficient $c = 12b(N)$ is the unique value for which the WDW ground-state energy reproduces the classical Havelock spectrum: if c were different, the quantum corrections to $V(\rho)$ would break the mode-independence of C_1 (the defining prop-

erty of the Havelock decomposition). In the classical limit $c \rightarrow \infty$: $V(\rho^*) = 0$, giving $C_1 = f(m^*, N)$ and hence $\Lambda = -4[f(m^*) - (N-1)]/[(N-1)\varepsilon^2]$.

Corollary 12.3 (The cosmological constant from the polygon number). *The sign of Λ is fixed by the polygon: $N \leq 6$: $f(m^*) < N-1$, so $\Lambda > 0$ (positive curvature, S^2 -like—surface curvature destabilises); $N = 7$: $f(3, 7) = 6 = N-1$, so $\Lambda = 0$ (flat); $N \geq 8$: $f(m^*) > N-1$, so $\Lambda < 0$ (negative curvature, $\mathbf{H}^2$ -like—surface curvature stabilises). The flat threshold $N = 7$ —Havelock’s 1931 stability boundary—sits exactly at $\Lambda = 0$.*

Remark 12.1 (Relation to Schur’s lemma). *In 2+1D, Step 4 of the proof is a special case of Schur’s lemma (isotropic sectional curvature implies constant curvature). The added value of the Havelock derivation is fourfold: (a) Steps 1–4 are new: polygon stability implies isotropic Green’s function, which Schur alone cannot provide; (b) the curvature value Λ is quantised by the polygon number N through the explicit formula $\Lambda = -4[f(m^*, N) - (N-1)]/[(N-1)\varepsilon^2]$; (c) the hypothesis of Schur’s lemma (isotropy) is derived from the Hadamard universality of the Green’s function, not assumed; (d) in $d \geq 4$, the Havelock condition (H1) provides a testable criterion (mode-independence of C_1) equivalent to the full Weyl-vanishing condition of Schur’s lemma, connecting the abstract curvature condition to the observable stability threshold.*

Remark 12.2 (Characterisation theorem in $d \geq 3$). *Steps 1–3 are dimension-independent: the csc^2 uniqueness forces a logarithmic interaction in any d (Step 1); the Hadamard parametrix $G(x, y) = U/\sigma^{(d-2)/2} + V \log \sigma + W$ contains $R(x)$ at order ε^2 in every d (Step 2); and (H3) forces $R = \text{const}$ (Step 3). Step 4 closes differently by dimension:*

Dimension	Weyl components	Status
$d = 2$ (2+1D)	0	Theorem: (H1) proved by the Havelock identity
$d = 3$ (3+1D)	10	Spatial equivalence: (H1) constrains traceless Ricci; full spacetime req
$d \geq 4$	≥ 35	Same, with more traceless Ricci components

In $d = 2$: the Havelock identity proves (H1) analytically—no assumption is needed. In $d \geq 3$: condition (H1) (mode-independence of C_1) is equivalent to the spatial Green’s function being isotropic, which is equivalent to maximal symmetry, which is equivalent to the vacuum Einstein equations with Λ . The soft Havelock decomposition (Proposition 9.2) shows that (H1) is approximately satisfied on any background, with residual $|R_m| < a|f_m|$ (verified for $N = 5, \dots, 20$ and $d = 3, 4, 5$): the violation of (H1) is controlled and measures the departure from Einstein geometry.

Supporting evidence: spectral focusing. By Theorem 9.3 and Corollary 9.4, $N_{\text{crit}}(\Delta)$ is non-increasing: once a polygon type becomes unstable, it remains unstable for all larger Δ .

This is the polygon analogue of the Raychaudhuri focusing theorem:

Concept	General relativity	Polygon system
Flow parameter	Affine parameter λ	Conformal dim Δ
Expansion	$\theta = \nabla \cdot k$ (decreasing)	$N_{\text{crit}}(\Delta)$ (decreasing)
Focusing	$d\theta/d\lambda \leq 0$	$dN_{\text{crit}}/d\Delta \leq 0$ (monotone)
Trapped	$\theta < 0$ (trapped surface)	$\lambda_m < 0$ (unstable mode)
Energy condition	$R_{\mu\nu}k^\mu k^\nu \geq 0$	$C_1(\xi) > 0$ (spectral gap)

The palindromic threshold as a Killing horizon. The WDW equation $[-1/(2c) \Psi'' + V(\rho) \Psi = 0]$ is the zero-energy Schrödinger equation in the potential V . The turning point $V(\rho^*) = 0$ is a *Killing horizon* in the minisuperspace geometry: the Euclidean continuation around ρ^* requires periodicity $\beta = 2\pi/|V'(\rho^*)| = 2\pi/\coth(\rho^*)$ to avoid a conical singularity [?]. This is the same calculation that gives the de Sitter temperature $T = H/(2\pi)$ —with the breathing mode ρ playing the role of the scale factor a . The Gibbons–Hawking temperature at the palindromic threshold is

$$T = \frac{V'(\rho^*)}{2\pi} = \frac{\coth(\rho^*)}{2\pi}. \quad (12.2)$$

Logical role of the Clausius relation. The vacuum Einstein equations $R = \text{const}$ are derived *independently* from entropy maximisation (Theorem 12.1, Steps 1–3). The Clausius relation at the Gibbons–Hawking horizon serves a *different* purpose: it extracts the gravitational coupling G_{eff} from the constant-curvature geometry. On a constant-curvature surface, there is one threshold (one N) and one temperature T . The Jacobson argument [??] then promotes G_{eff} to the matter coupling: $\delta Q = T \delta S$ at the Killing horizon gives the full Einstein equations $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G_{\text{eff}} T_{\mu\nu}$.

The three thermodynamic quantities are determined by the BO potential $V(\rho) = \log(2 \sinh \rho) + b(N) - f(m^*, N)$:

- $T = V'(\rho^*)/(2\pi) = \coth(\rho^*)/(2\pi)$ (Unruh temperature from the surface gravity $\kappa = V'$);
- $\delta Q = \kappa \times A \times \delta \rho = \coth(\rho^*) \times 2\pi \sinh(\rho^*) \times \delta \rho$ (surface gravity $\times$ horizon circumference $\times$ displacement);
- $\delta S = (dS/d\rho) \times A \times \delta \rho$ where $dS/d\rho = \coth(\rho^*)/2$ is the entropy production *per unit length* from the single zero mode ($S = -\frac{1}{2} \log |\lambda_{m^*}|$, $dS/d\rho = \coth(\rho^*)/2$ in the regularised limit), and $A = 2\pi \sinh(\rho^*)$ is the horizon circumference.

The common factor $\coth(\rho^*) = V'(\rho^*)$ is *not* a free choice: it is determined by the csc^2 kernel ($V = \log(2 \sinh \rho)$ is the $\mathbf{H}^2$ Green’s function). The Clausius ratio is

$$\frac{\delta Q}{T \delta S} = \frac{4\pi}{\kappa} = 4\pi \tanh(\rho^*) = 4\pi - O(e^{-2\rho^*}). \quad (12.3)$$

The Jacobson identification $\delta Q/(T \delta S) = 1/(4G)$ gives $G_{\text{eff}} = 1/(16\pi \tanh(\rho^*)) \rightarrow 1/(16\pi)$ as $\rho^* \rightarrow \infty$ (equivalently, $8\pi G_{\text{eff}} = 1/(2 \tanh(\rho^*)) \rightarrow 1/2$ in natural units). The ratio $4\pi \tanh(\rho^*)$ is an algebraic identity in $\coth(\rho^*)$, holding exactly at every palindromic threshold. The ? entanglement argument: positivity of relative entropy $S(\rho||\sigma) \geq 0$ in the unitary boundary CFT implies $\delta Q = T \delta S$ at every Rindler horizon, The Clausius relation is verified numerically at all palindromic thresholds $N = 7, \dots, 17$ (test suite: `test_jacobson_derivation.py`). The extension to the full matter-coupled Einstein equations $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}$ requires the Jacobson thermodynamic argument (?: ?): $\delta Q = T \delta S$ at every local Rindler horizon implies Einstein with matter. The identification of palindromic thresholds as Rindler horizons uses the modular Hamiltonian of the angular bath, which equals the Rindler energy by the Bisognano–Wichmann theorem for the boundary CFT (Paper IV, Theorem 20.3).

Matter coupling. On a quotient $\Gamma \backslash \mathbf{H}^2$, the Selberg correction δC_1 shifts the threshold $\rho^* \rightarrow \rho_{\text{matter}}^*$, but the Clausius ratio retains the form $4\pi \tanh(\rho_{\text{matter}}^*)$ (verified on the Bolza surface with $|\delta C_1| \leq 0.021$). The ratio 4π is a local geometric identity (the metric is locally $\mathbf{H}^2$), giving $G_{\text{eff}} = 1/(16\pi)$ in natural units ($8\pi G_{\text{eff}} = 1/2$ as $\rho^* \rightarrow \infty$).

The arithmetic of the palindromic hierarchy is the interference between conformal invariance and geometric curvature—the residue that the renormalisation group cannot wash away.

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